Luke Johnson

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EDUCATION

University of Washington, Information School — Seattle, WA
Bachelor of Science in Informatics — June 2026
Relevant Coursework: Database & Data Modeling, Product and Information Systems Management, Enterprise Risk Management, Python with Applied Machine Learning
Activities: Product Management Club, Applied Analytics Club

EXPERIENCE

Americans4Equality (A4E) — Bellevue, WA
Program Management Intern — August 2026 – October 2026

- Rebuilt nonprofit's website from zero to production as **PM and lead developer** using React 19, TypeScript, Vite, and Cloudflare Pages via GitHub Actions CI/CD; architected centralized TypeScript content management enabling non-technical staff to update all copy without touching component code.
- Developed **multi-platform social media strategy** across Instagram, TikTok, YouTube, and LinkedIn; introduced Claude-powered AI content workflows to accelerate production; built **JotForm student intake and screening system** to evaluate program eligibility at scale; managed all workstreams in Monday.com and Office 365.

Extern — Remote
BeReal Product Management Externship — June 2026 – August 2026

- Identified highest-priority MAU growth feature from 6 candidates, as measured by **RICE score 12.8 for "Buddy Nudge"**, by mapping a 5-stage user journey (Awareness through Retention), analyzing 4 core retention mechanisms, and benchmarking BeReal against TikTok, Instagram, and Snapchat on DAU/MAU and feature differentiation.
- Wrote a formal **PM problem statement** identifying forced video adoption as the primary friction disrupting BeReal's authenticity-first value proposition; validated feature through structured focus group testing; delivered **stakeholder pitch with business rationale and implementation roadmap**.

Extern — Remote
Amazon Operational Strategy and People Analytics Externship — December 2025 – February 2026

- Surfaced **10–15% operational inefficiencies** invisible in existing reporting, projecting an **8% reduction in downtime**, by building an analytical framework from scratch using Python and SQL; produced severity-ranked findings report now under org-wide review.
- Translated complex technical findings into **executive-ready recommendations** for senior leadership; leveraged **AI tools daily** to accelerate analysis while maintaining full ownership of every interpretation and recommendation.

PROJECTS

Wavelength – PM Lead, AI-Native Mobile App — March 2026 – Present
PM Lead + Developer | React Native, TypeScript, Supabase, Claude API, Figma, Jira, GitHub

- Led a 5-person cross-functional team toward App Store launch as PM lead; owned **sprint planning, roadmap reviews, backlog grooming, and acceptance criteria**; integrated **Claude API as the core AI matching layer** powering personalized recommendations; grounded every feature in a specific customer problem before writing a single requirement.
- Defined the metric each feature should move before shipping; monitored **post-launch data to validate behavior change**; applied AI tools to generate first-draft requirements then critically reviewed every output; built direct working relationships with engineering and design partners aligning early.

Spec Writer – AI-Powered PM Productivity Tool — August 2026 – Present
Developer | React 18, Vite, Tailwind CSS, FastAPI, Claude API (tool-use / structured output), GitHub Actions

- Built a full-stack tool that turns raw meeting notes, Slack threads, and bullet points into a **structured PRD** using Claude API **tool-use with forced JSON schema output** (submit_prd and submit_critique tools) so responses are always parseable — never free text requiring hand-parsing; stateless FastAPI backend with Vite dev proxy eliminates CORS configuration.
- Built "Poke Holes" mode that red-teams the generated spec the way a senior PM would, surfacing **weak assumptions, missing edge cases, and scope creep** before an engineer raises them in review; added clarifying questions mode that fires when input lacks a clear user or problem instead of hallucinating spec details.

Autonomous Outage Response System – Agentic AI Incident Pipeline — August 2026 – Present
Developer | Python, FastAPI, SQLite, Claude API (tool-use / function calling), Pydantic, pytest

- Built an autonomous incident response pipeline triggered by a webhook: identifies the likely bad commit via **git blame scoring**, retrieves best-match runbook via **TF-IDF similarity search**, estimates user impact using **Claude tool-use and function calling** over live metrics, and posts a structured Slack brief to the on-call channel automatically.
- Designed for production reliability: offline-first with **29 fully offline tests** and deterministic fallbacks so the suite runs free without network; best-effort pipeline where a failing phase produces a partial brief rather than crashing; auto-generates a **blameless postmortem on incident resolution** using Claude with FastAPI REST API and 6 endpoints..

AI Prompt Injection Cybersecurity Research – LLM Security & Applied AI Research — Sept – Dec 2025
Researcher (team of 4) | Python, Claude API, ChatGPT API, GitHub Copilot, GitHub

- Synthesized findings across 20 academic sources and 4 enterprise platforms including Microsoft Copilot Studio into **executive-ready recommendations** mapped to OWASP LLM Top 10 and NIST AI RMF; evaluated **Claude API and ChatGPT API behavior** for reliability, edge cases, and failure modes across agentic systems.

LEADERSHIP & COMMUNITY INVOLVEMENT

Pi Kappa Alpha Fraternity — Seattle, WA
Internal VP | Secretary | Recruitment Chairman | Events Chairman — September 2022 – June 2026

- Raised over **$28,000** for Seattle Children's Hospital and improved alumni engagement **27%**; demonstrated ownership, stakeholder management, and cross-functional collaboration across 4 elected leadership roles serving 60+ members over 4 years.
- Coordinated large-scale recruitment campaigns and events reaching 500+ attendees; managed vendor relationships, budgets, and logistics across concurrent workstreams.

ADDITIONAL INFO/TECHNICAL SKILLS

Programming Languages: Python, JavaScript, R, SQL, HTML, CSS, React, and Node.js
AI and Data: LLM security and prompt injection research, prompt engineering for defense strategies, data modeling, and analytics.
Platforms and tools: Firebase, GitHub, Azure DevOps, Figma, Miro, ChatGPT, Personal LLM, Claude
Core skills: Leadership, Cross-Functional Collaboration, Scrum methodology, user research, quality assurance, analytical thinking
Languages: English (Fluent), Vietnamese (Native), Spanish (Intermediate)